

Lecture 15: Point Estimates and Sampling Variability

Terminology: Parameter, Statistic, Point Estimate

Sampling Variability

Example: Heights of men at UNC

70, 65, 69, 69, 75, 74	$\bar{x} = 70.3$
68, 67, 68, 70, 64, 68	$\bar{x} = 67.5$
72, 71, 71, 74, 69, 72	$\bar{x} = 71.5$
70, 65, 69, 69, 75, 74	$\bar{x} = 70.3$
74, 69, 64, 64, 66, 76	$\bar{x} = 68.8$

Here are the values of $\bar{x}$ for 5 samples of size 100 from the same population:

70.0, 69.4, 68.9, 69.1, 68.9

Young, Underemployed and Optimistic

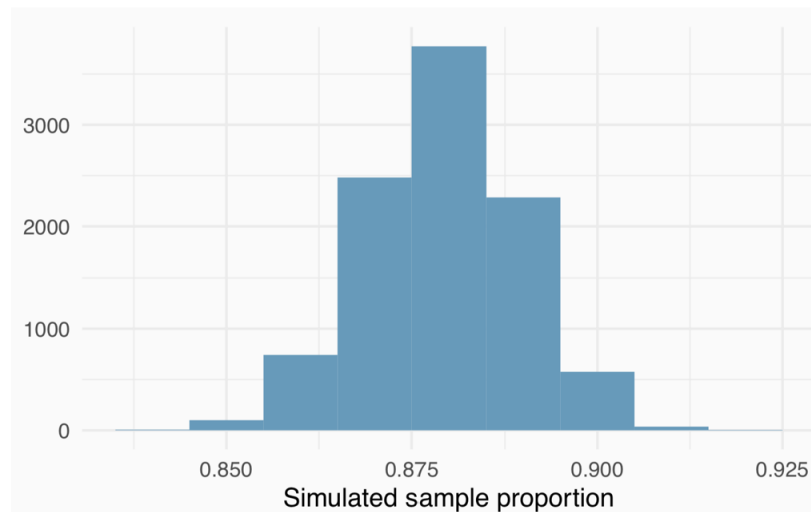
Coming of Age, Slowly, in a Tough Economy

Young adults hit hard by the recession. A plurality of the public (41%) believes young adults, rather than middle-aged or older adults, are having the toughest time in today's economy. An analysis of government economic data suggests that this perception is correct. The recent indicators on the nation's labor market show a decline in the **Tough economic times altering young adults' daily lives, long-term plans.** While negative trends in the labor market have been felt most acutely by the youngest workers, many adults in their late 20s and early 30s have also felt the impact of the weak economy. Among all 18- to 34-year-olds, fully half (49%) say they have taken a job they didn't want just to pay the bills, with 24% saying they have taken an unpaid job to gain work experience. And more than one-third (35%) say that, as a result of the poor economy, they have gone back to school. Their personal lives have also been affected: 31% have postponed either getting married or having a baby (22% say they have postponed having a baby and 20% have put off getting married). One-in-four (24%) say they have moved back in with their parents after living on their own.

The general public survey is based on telephone interviews conducted Dec. 6-19, 2011, with a nationally representative sample of 2,048 adults ages 18 and older living in the continental United States, including an oversample of 346 adults ages 18 to 34. A total of 769 interviews were completed with respondents contacted by landline telephone and 1,279 with those contacted on their cellular phone. Data are weighted to produce a final sample that is representative of the general population of adults in the continental United States. Survey interviews were conducted under the direction of Princeton Survey Research Associates International, in English and Spanish. Margin of sampling error is plus or minus 2.9 percentage points for results based on the total sample and 4.4 percentage points for adults ages 18-34 at the 95% confidence level.

Example: Suppose the proportion of all American adults who support the expansion of solar energy is $p = 0.88$, but we don't have access to the population of all American adults (a quite likely scenario) and thus can't know this true value of p .

If we are interested in estimating p , a reasonable approach might be:



We will never actually observe a **sampling distribution** like this one, but the **Central Limit Theorem** describes its shape.

Derivation of the sampling distribution for a sample proportion $\hat{p}$

Necessary conditions for CLT to apply:

- Independent Observations

- Adequate Sample Size

Example: Randomly sample $n = 2500$ American adults and ask whether they like “The Office”. 2400 respond “Yes” and 100 respond “No”.

If the true proportion p of American adults who like “The Office” was 0.9, with what probability would we obtain a random sample of 2500 adults with at least 2400 who like the show?

More on CLT conditions:

- When either np or $n(1 - p)$ is small, the sampling distribution is more discrete and more skewed. The larger both np and $n(1 - p)$ are, the more normal the distribution looks.
- When np and $n(1 - p)$ are both very large, the discreteness of the sampling distribution is hardly evident, and the distribution looks much more like a normal distribution.

Example: Suppose that only 15% of Americans can explain what “data science” is.

- If we ask a random sample of 75 Americans to explain what data science is, what is the probability that at least 20% of them can do it?

- If we ask a random sample of 750 Americans to explain what data science is, what is the probability that at least 20% of them can do it?

The strategy of using a sample statistic to estimate a population parameter is quite common, and it's a strategy that we can apply to other statistics besides proportions, e.g.:

- Take a random sample of students at UNC and ask them how many extracurricular activities they are involved in to estimate the average number of extracurriculars across all UNC students.
- Also take a random sample of Duke students and record numbers of extracurricular activities, compare sample means to gauge potential differences between UNC and Duke student bodies.